# Diego Macías Vega

Final-year science student in Madrid · physics, AI and engineering

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## PROFILE

I direct software projects built on physics and mathematics, written with AI coding assistants, and check them against measurement: reproduced baselines, validation suites and published negative results.

## EDUCATION

**Bachillerato in Sciences and Technology** IES San Agustín del Guadalix, Madrid — 2025 – 2027

- Mathematics II, Physics, Technology and Engineering, Computer Science. Final year; graduating June 2027.

**University applications, September 2027 entry** the Netherlands — in preparation

- Computer Science and Engineering (TU Delft, selection test) · Applied Physics (TU Eindhoven) · Advanced Technology (University of Twente).

**CS50x: Introduction to Computer Science** Harvard University, online — Started 2026

## PROJECTS

**RAAN · Open Orbital Intelligence Lab** Python, TypeScript, WebGL, LightGBM — 2026

- Reproducible lab on public orbital data. Digital twin of 20,152 catalogued objects propagated with SGP4; propagator checked against Vallado's 666 reference vectors (max difference 3.5e-5 km).
- ESA Collision Avoidance Challenge: naive forecast reproduced exactly (L = 0.694). Best model scores 0.627 on the test set but fails cross-validation; reported as a negative result.
- Manoeuvre detection on 15 satellites, leave-one-satellite-out: F1 0.81 (classifier), 0.74 (physics baseline), 0.09 (chance). 6-page technical report.
- Code mostly written by an AI coding assistant and reviewed by an independent session; I set the question and the rules and answer for every claim.

**Syndrome Lab** quantum error correction — 2026, in progress

- Neural decoders vs minimum-weight perfect matching on the surface code, built with Stim and PyMatching. Results and report to be published late November 2026.

**gravity-lab** C++17, numerical methods — 2026

- N-body simulator of the Solar System validated against measurement: reproduces Mercury's relativistic perihelion precession (42.81 arcsec/century; accepted 42.98, error 0.4 %).
- Four integrators compared; Yoshida 4 (symplectic) holds energy error at 1.4e-11 over 100 years. 31 validation cases in CI. Browser build online.

**KAIROS** Python, FastAPI, PostgreSQL + pgvector, Docker — 2026

- Local-first personal AI assistant: 17 agents, 300+ tests, sandboxed self-modification that never merges without my approval. Architecture and decisions mine (77 decision records); code by Claude.

**Other** — 2026

- Chess trainer (Stockfish and Maia, explanations fact-checked by test) · Quant Path (drills and 110 problems for quantitative trading, 101 verified by code) · TraceGuard (firmware changes checked against safety rules; paused MVP) · Sympto+ (multilingual health-information web app; paused).

## EXPERIENCE

**Private tutor** Madrid — ongoing

- One-to-one tuition for school students.

## SKILLS, LANGUAGES AND OTHER

**How I build:** I design, direct, review and test projects built with AI coding assistants (Claude Code). I define the problem, the evaluation and the acceptance criteria, check results against references and decide what ships. I cannot yet write these codebases from scratch; I am learning programming fundamentals (Python, algorithms) alongside.

**Working knowledge:** Python (beginner, learning), Git and GitHub, Linux/WSL, reading and testing code. Used in my AI-assisted projects, not written by me: C++17, TypeScript, FastAPI, PostgreSQL, Docker.

**Concepts I can explain:** orbital propagation (SGP4), baselines, cross-validation and bootstrap intervals, gradient boosting at a conceptual level, the surface code and minimum-weight perfect matching.

**Languages:** Spanish (native) · English, Cambridge C1 Advanced (score 186, obtained at 16) · Chinese, YCT 2.

**Interests:** superintelligence and robotics · physics, quantum above all · machine learning and AI · mathematics. **Sport:** judo, black belt.